

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

Generated on: 2026-01-04 17:11:54

Total Marks: 100

Examination Paper

Question Q1 (20 marks)

- a) Draw an EER diagram for the following scenario: The University has several departments such as Computer Science, Mathematics and Physics. Each department has a name and a head of department. Students are enrolled in courses offered by these departments. Courses have a course number, name, credits, and semester. Students register for courses and pay tuition fees. Members can borrow books from the library. The university has several sections such as Computer Science, Mathematics and Physics. Each section has a unique identifier and is headed by a head librarian. Members can put hold requests on books in these sections. Librarians have names, IDs, phone numbers, and addresses. (5)
- b) Explain the EER model concepts of specialization and generalization, and provide an example of how they can be applied to the University scenario. Be sure to include a diagram showing the relationships between entities in your answer. (6)
- c) Describe the constraints imposed by overlapping and total ISA relationships on the University EER model. Provide examples of how these constraints can be used to ensure data consistency and integrity. (4)
- d) Design a relational schema for the University database based on the EER model, including tables for departments, courses, students, members, sections, and librarians. Be sure to include primary keys, foreign keys, and integrity constraints in your design. (5)

Question Q2 (20 marks)

- a) Draw an ER diagram for the scenario below. A University has several departments such as Computer Science, Mathematics and Physics. Each department has a name and a head of department. Students belong to a department and have a roll number and name. (3)
- b) List the types of relationships that can exist between entities in an ER diagram. Provide examples for each type. (2)
- c) Explain the concept of weak entity and how it is identified in an ER model. (1)
- d) List three common attributes that can be used to define a composite attribute in an ER diagram. (2)
- e) Explain the concept of ternary relationship and how it is represented in an ER model. (2)

- f) Draw a binary relationship between two entities, where one entity has multiple instances of another entity. Explain the cardinality ratio used. (2)
- g) Explain how to identify candidate keys in an ER diagram. (2)
- h) List the types of relationships that can exist between entities in a binary relationship. Provide examples for each type. (2)
- i) Construct an ER diagram for the scenario below: A Library has several sections such as fiction, children and technology. Each section has a name and a number(unique) and its headed by a head librarian. Books belong to a section and have a title, authors, ISBN, year and a publisher. Members can borrow books and have a membership number, an address and a phone number. (4)

Question Q3 (22 marks)

- a) Draw the EER diagram for the given scenario and label all entities, attributes, and relationships. Explain the meaning of each entity type and its relationships with other entity types. (2)
- b) Explain the concept of specialization in the EER model, providing an example of how it can be applied to the given scenario. Discuss the advantages and disadvantages of this approach. (2)
- c) Analyze the ISA relationships between entity types in the given scenario. Justify why certain relationships are chosen over others, considering the context of the database application. (2)
- d) Design a trigger to enforce the constraint that an employee cannot have a salary greater than their manager's salary in the given scenario. Explain the logic behind the trigger and its implementation. (3)
- e) Map the given ER diagram to a relational model, considering the constraints and relationships between entity types. Identify any potential issues or challenges in this mapping process. (2)
- f) Explain how the EER model can be used to represent the given scenario's complex relationships and constraints, including overlapping, disjoint, and total relationships. (3)
- g) Calculate the number of possible values for a certain attribute in the given scenario, considering the constraints and relationships between entity types. (2)
- h) Design a database schema to store the data from the given scenario, including tables, columns, and primary keys. Explain the reasoning behind your design choices. (3)

- i) Implement a stored procedure to perform a complex query on the database schema designed in Part e.iv, considering the constraints and relationships between entity types. (3)

Question Q4 (20 marks)

- a) Describe the process of logical database design and its importance in the database design process. (5 marks) (3)
- b) Explain the relational model and its components, including relations, attributes, and keys. Provide an example of a relation schema. (5 marks) (3)
- c) Design an ER diagram for a university database that includes departments, students, and faculty members. Ensure the diagram accurately represents the relationships between entities. (5 marks) (1)
- d) Explain the concept of normalization and its benefits in database design. Provide examples of different normal forms and their applications. (3 marks) (3)
- e) Design a relational schema for a university database that includes departments, students, and faculty members. Ensure the schema accurately represents the relationships between entities. (3 marks) (3)
- f) Explain the concept of data integrity and its importance in database design. Provide examples of different types of integrity constraints. (4 marks) (4)
- g) Design a query to retrieve all students who are enrolled in a specific department, along with their names and grades. Ensure the query accurately represents the required information. (3 marks) (3)

Question Q5 (18 marks)

- a) Draw an EER diagram for the given scenario and label each entity, attribute, and relationship. Explain the purpose of each entity, attribute, and relationship in 50 words or less. (5)
- b) Explain the concept of specialization and generalization in the context of EER models. Provide an example of how these concepts can be applied to a real-world scenario, such as a university's departmental structure. (2)
- c) Describe the ISA relationship between 'Employee' and 'Permanent Employee'. How does this relationship ensure data consistency and integrity in a database? (2)
- d) Design a trigger to enforce the constraint that an employee cannot work in more than

two departments. Explain how the trigger will prevent duplicate entries. (2)

e) Consider the following table: Transaction(tid, accountNo, type, amount, date). Write a trigger to ensure that no customer withdraws more than the given limit of Rs. 40,000 per day. (1)

f) Create an assertion to enforce the constraint that a student's salary cannot exceed their manager's salary. Explain how the assertion will ensure data consistency and integrity. (1)

g) Map the following ER diagram to a relational model: [PLACEHOLDER FIGURE] (Description of what diagram should show). (2)

h) Explain how the EER model's specialization and generalization concepts can be used to represent a university's departmental structure. Provide an example of how these concepts can be applied in practice. (2)

i) Describe the differences between a domain constraint, referential integrity constraint, and key constraint in the context of database design. Provide examples of each type of constraint. (1)